Human Health & Diseases

Health can be defined as a state of complete physical, mental, and social (emotional fitness) well-being. It increases longevity of people and reduces infant and maternal mortality rates.

- Balanced diet, personal hygiene, and regular exercise. Yoga has been practised since time immemorial to achieve physical & mental health.
- Awareness about diseases and their effect on different bodily functions.
- Vaccination against infectious diseases.
- Proper disposal of wastes.
- Control of vectors.
- Maintenance of hygiene in food and water resources.

*Health is more than just the absence of disease!

Infectious diseases: Diseases which are easily transferred from one person to another. (E.g., AIDS, Common cold).

Non-infectious diseases: Diseases which are not easily transferred from one person to another. (E.g., Cancer).

Common diseases in Humans :—

Pathogens: Disease-causing organisms (bacteria, fungi, viruses, protozoans, helminths). Most parasites are pathogens as they cause harm by living in or on the host. Pathogens must adapt to the host's environment (e.g, gut pathogens survive low stomach pH and resist digestive enzymes).

* Typhoid :—

Caused by: Bacteria — Salmonella.

Attacks small intestine via contaminated food/water.

Symptoms: High fever (39-40°C), weakness, stomach pain, constipation, headache, loss of appetite. Can cause death in severe cases.

Note: Detected through Widal test.

Trivia: Mary Mallon (Typhoid Mary) was a carrier cook who spread it through food.

* Pneumonia :—

Caused by: Bacteria — Streptococus pneumoniae, Haemophilus influenzae.

Attacks alveoli (they get filled with fluid).

Symptoms: Fever, chills, cough, headache, lips/fingernails may turn grey to bluish.

Acquired by inhaling droplets/aerosols or sharing glasses/utensils.

* Common cold :—

Caused by: Virus — Rhino virus.

Attacks nose & respiratory tract (not lungs).

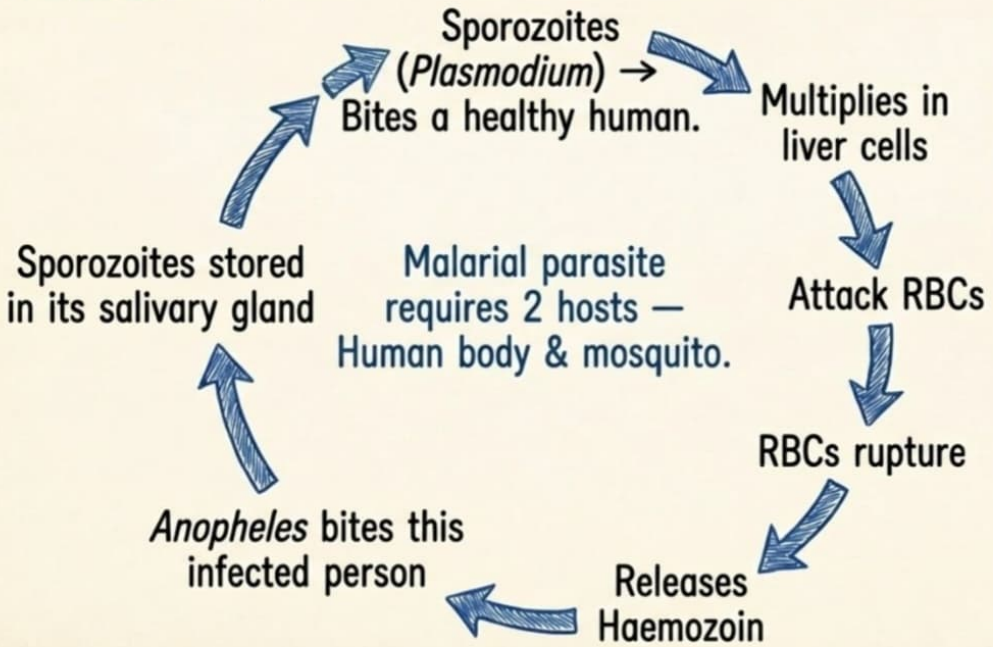
Symptoms: Nasal congestion/discharge, sore throat, hoarseness, cough, headache, tiredness (lasts 3-7 days).

Acquired via infected cough/sneezes or contaminated objects (books, pens, doorknobs, keyboards).

*Dysentery, plague, diphtheria are also bacterial diseases.

* Malaria :—

Caused by protozoan — *Plasmodium* (*P. vivax*, *P. malaria*, *P. falciparum* — deadliest) via female *Anopheles* mosquito.



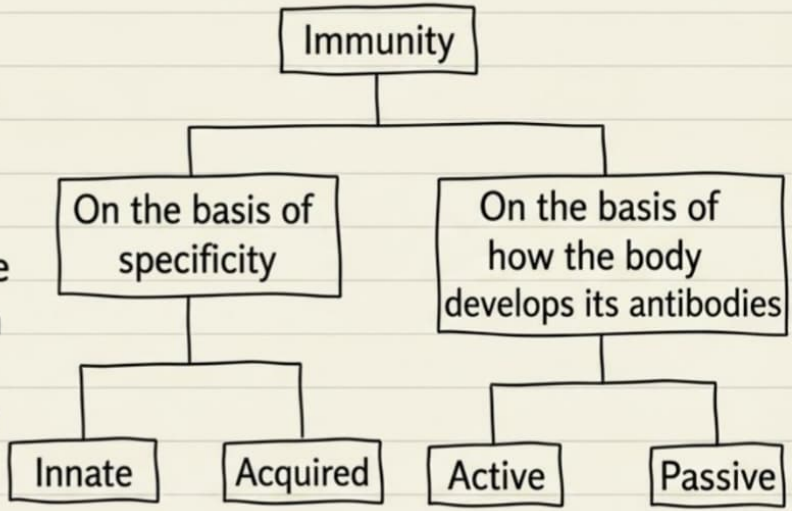
*Haemozoin is the toxic substance responsible for the recurring chills and fever for 3-4 days!

* Prevention :—

- ☐ Maintenance of personal hygiene.
- ☐ Public hygiene: proper waste/excreta disposal, periodic cleaning/disinfection of water reservoirs, pools, tanks.
- ☐ Air-borne diseases: avoid close contact with infected persons.
- ☐ Vector diseases: avoid water stagnation, clean household coolers, use mosquito nets, introduce Gambusia fish (feeds on mosquito larvae), spray insecticides in drainage/ditches. Doors/ windows need wire mesh.

Immunity :—

The ability of the host to fight the disease-causing organisms.



* Amoebiasis (Amoebic dysentery) :–

Caused by: Protozoan – *Entamoeba histolytica* (via houseflies).

Attacks large intestine.

Symptoms: Constipation, abdominal pain, cramps, stools with excess mucous and blood clots.

Acquired by having contaminated food and drinking water.

* Ascariasis :–

Caused by: Helminths – *Ascaris* (roundworm).

Attacks Intestine.

Symptoms: Internal bleeding, muscular pain, fever, anaemia, blockage of intestinal passage.

Acquired via contaminated water, vegetables, fruits.

Eggs are released in faeces infecting soil/water.

* Filariasis or Elephantiasis :–

Caused by: Wuchereria (filarial worm) (W. bancrofti, W. malayi).

Attacks lymphatic vessels of lower limbs & genital organs.

Symptoms: Slow chronic inflammation of organs.

Acquired by bite of infected female Culex mosquito vector.

* Ringworms :–

Caused by: Fungi – Microsporum, Trichophyton, Epidermophyton.

Attacks skin, nails, and scalp.

Symptoms: Dry, scaly lesions with intense itching. Grows in groin/between toes due to heat & moisture.

Acquired from soil, towels, clothes, or comb of infected person.

* Innate immunity :—

Non-specific defense, present at birth.
Attained by providing barriers to foreign agents.

1. **Physical barriers:** Skin prevents micro-organism entry. Mucus coating lining respiratory, gastrointestinal, and urogenital tracts trap microbes.

2. **Physiological barriers:** Acid in stomach, saliva in mouth, tears from eyes — prevent microbial growth.

3. **Cellular barriers:** Leukocytes (WBCs) like PMNL-neutrophils, monocytes, and natural killer lymphocytes in blood + macrophages in tissues phagocytose and destroy microbes.

4. **Cytokine barriers:** Virus-infected cells secrete interferons to protect non-infected cells.

* Acquired immunity :—

- Pathogen-specific, characterised by memory.
- Primary response is low intensity; subsequent encounter elicits a highly intensified secondary/anamnestic response.
- Carried out by B-lymphocytes & T-lymphocytes.
- Humoral response: B-lymphocytes produce protein antibodies in blood. T-cells help them. Each antibody has 4 peptide chains (2 light, 2 heavy). Represented as H2L2. E.g.: IgA, IgM, IgE, IgG.
- Cell-mediated (CMI): Mediated by T-lymphocytes. Body differentiates self/non-self. CMI is responsible for graft rejection (tissue/blood matching essential before transplants).

BASIS	TYPE OF IMMUNITY	METHOD OF PROTECTION	EXPLANATION
On the basis of specificity	Innate	Physical Barrier, Physiological Barrier, Cellular Barrier, Cytokine Barrier	Skin/mucus trap microbes. Acid/saliva/tears prevent growth. PMNL/macrophages destroy microbes. Interferons protect non-infected cells.
"	Acquired	Humoral Immune Response, Cell mediated immune response	B-cells produce proteins, antibody has 4 peptide chains (H_2L2). T-cells mediate CMI, responsible for graft rejection.
On the basis of how the body develops its antibodies	Active	Infectious organisms or microbes are injected	Consists of producing antibodies on attack of an antigen. To fight these, infectious organisms are injected during natural infection.
"	Passive	Ready-made antibodies are directly given	Yellowish fluid called colostrum secreted by mother has abundant IgA. Foetus also receives antibodies through placenta.

* Auto immunity :-

Memory-based acquired immunity evolved in higher vertebrates based on the ability to differentiate foreign organisms from self-cells.

Corollaries:

1) Higher vertebrates can distinguish foreign molecules & organisms.

2) Sometimes, due to genetic & other unknown reasons, the body attacks self cells. This leads to damage in body and is called an auto immune disease.

E.g.: Rheumatoid arthritis.

* Allergies :-

The exaggerated response of the immune system to certain environmental antigens (allergens).

Antibodies produced: IgE type.

Examples: pollen grains, dust mites, animal dander.

Symptoms: sneezing, watery eyes, running nose, difficulty breathing.

Occurs due to release of histamine & serotonin.

Diagnosis/Treatment: Patient exposed to very small doses of allergens to determine cause. Drugs like anti-histamine, adrenaline, & steroids quickly reduce symptoms.

* Active immunity :-

Host is exposed to antigens (living/dead microbes or proteins), producing antibodies. It is slow and takes time to give its full effective response. Induced by injecting microbes during immunisation or gaining access during natural infection.

* Passive immunity :-

Readymade antibodies are directly given to protect against foreign agents. E.g., Colostrum (IgA) during lactation, or placental transfer to foetus.

Ways to boost immunity :

Vaccination

Antigenic proteins or inactivated/weakened pathogens (vaccine) introduced into the body. Generates memory-B & T-cells that recognize pathogens quickly on subsequent exposure and overwhelm them with antibodies. Neutralizes agents during actual infection.

Immunisation

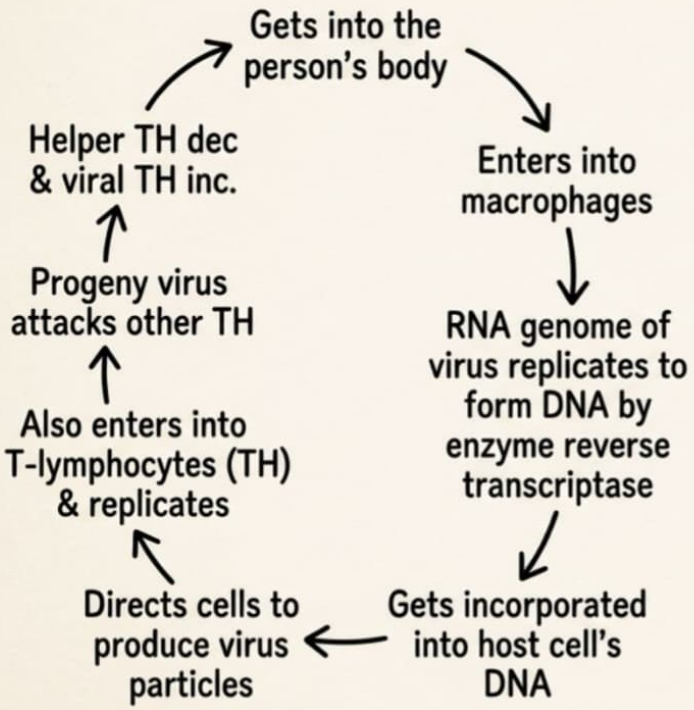
Direct injection of preformed antibodies/antitoxins for quick immune response. E.g., Tetanus, or preformed antibodies against snake venom (passive immunisation).

*Hepatitis-B vaccine is produced by yeast via recombinant DNA technology, allowing large-scale production.

Immune system consists of lymphoid organs, tissue, cells & soluble molecules (antibodies).

TYPE OF LYMPHOID ORGAN	FUNCTION	ORGANS (Shape / Location / Function)
Primary Lymphoid Organs	*Immature lymphocytes differentiate into antigen-sensitive lymphocytes. Provide micro-environment for T-lymphocytes.*	• ***Bone marrow***: *Spongy bone | Center of bone | Main organ where all blood cells including lymphocytes are produced.* • ***Thymus***: *Bilobed | Near heart, beneath breastbone | Quite large at birth, keeps reducing with age, very small by puberty.*
Secondary Lymphoid Organs	*After maturation, lymphocytes migrate here.*	• ***Spleen***: *Bean shaped | Upper left abdomen | Contains lymphocytes/phagocytes. Blood filter (traps micro-organisms). Large reservoir of erythrocytes (RBCs).* • ***Lymph nodes***: *Solid structures | Various lymphatic locations | Traps antigens/microbes, activates lymphocytes, causes immune response.* • ***Lymphoid tissues***: *Tissues | Lining of major tracts (resp, digest, urogenital) | Mucosa Associated Lymphoid Tissue (MALT). Constitutes ~50% of lymphoid tissue.*

* Lifecycle :—



* Pathology & Testing :

- Decrease in TH leads to bouts of fever, diarrhoea, weight loss (immuno-deficient).
- Tested via ELISA (Enzyme Linked Immuno-Sorbent Assay).
- Treated partially by anti-retroviral drugs.

* Prevention & Govt Steps :

- No cure; conscious behaviour prevention is best. Regular checkups, safe sex/condoms.
- NACO & NGOs spread awareness. WHO programs.
- Steps taken: Making blood banks safe from HIV infection, ensuring use of disposable needles in hospitals, free condom distribution, controlling drug abuse.

AIDS

Acquired Immuno Deficiency Syndrome.
Acquired during lifetime (not a congenital disease).
Syndrome = group of symptoms.
First reported in 1981.

Cause →

HIV (Human Immuno Deficiency) virus.
A member of the retrovirus group with an envelop enclosing an RNA genome.

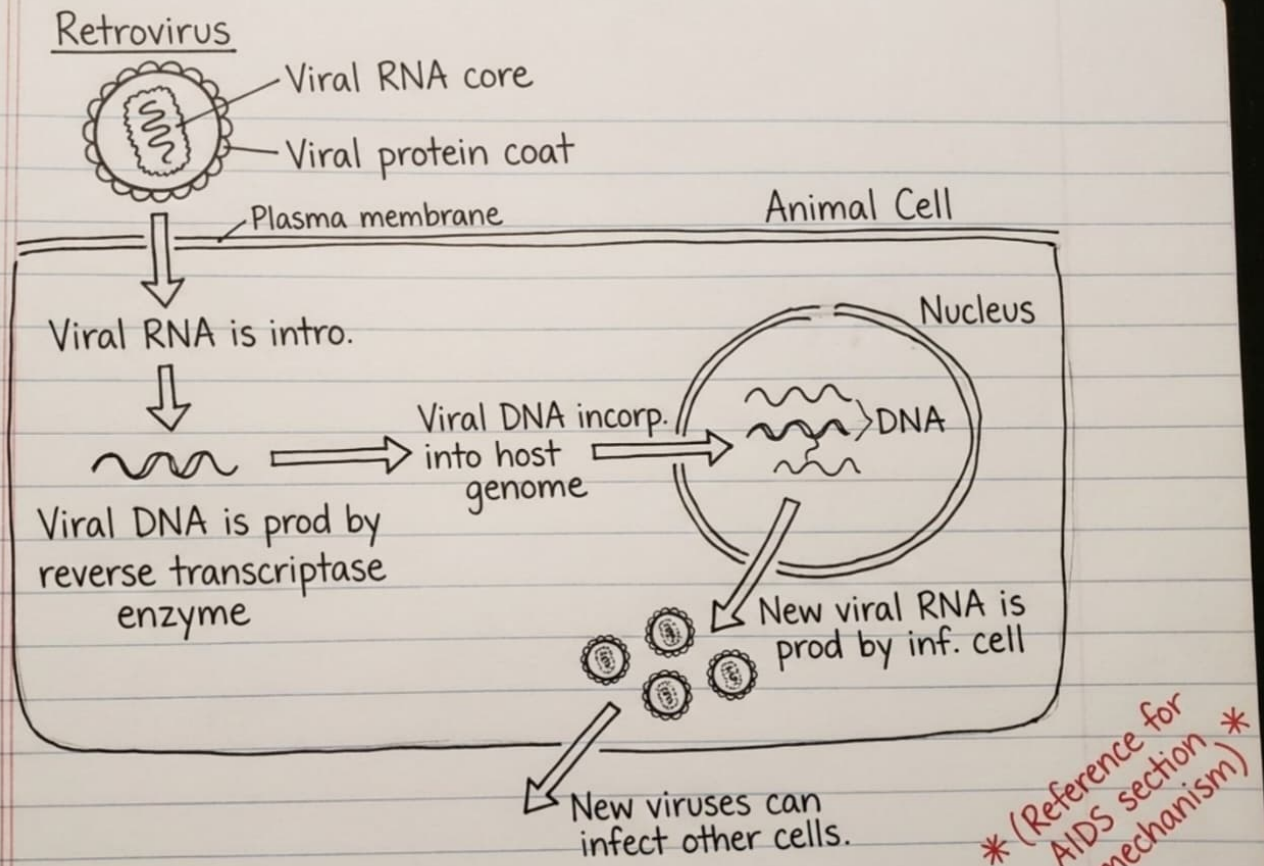
Transmission →

- Sexual contact with infected person.
- Transfusion of contaminated blood & products.
- Sharing infected needles (intravenous drug abusers).
- Infected mother to child through placenta or breastfeeding.

Not spread by mere touch! Only via body fluids.

High Risk Individuals →

- Individuals with multiple sexual partners.
- Intravenous drug addicts.
- Individuals requiring repeated blood transfusions.
- Children born to HIV infected mothers.


Retrovirus
Viral RNA core
Viral protein coat
Plasma membrane
Animal Cell
Viral RNA is intro.
Nucleus
Viral DNA incorp. into host genome
DNA
Viral DNA is prod by reverse transcriptase enzyme
New viral RNA is prod by inf. cell
New viruses can infect other cells.
* (Reference for AIDS section mechanism) *

Cancer :—

One of the most dreaded diseases of human beings and is a major cause of death all over the globe.

* Life cycle :—

Normal cells have a property called 'contact inhibition' (contact with other cells inhibits growth).

Cancer cells lack this property, continuously dividing to form a mass of cells called a tumour.

Benign tumors:	Malignant tumors:
Remain confined to their location & don't spread to other parts of the body.	Mass of neoplastic cells. Grow rapidly, invading & damaging normal tissue. Compete for vital nutrients. They reach different sites through blood and start a new tumour (Metastasis).

* Causes (Carcinogens) :—

Normal cells change into neoplastic cells by physical, chemical & biological agents.

1. Physical: Ionising radiations (X-rays, Y-rays) & Non-ionising radiations (UV rays) cause DNA damage leading to neoplastic transformation.

2. Chemical: Chemical carcinogens present in tobacco lead to lung cancer.

3. Biological: Cancer causing viruses (oncogenic viruses) have viral oncogenes. Cellular/proto oncogenes in normal cells can be activated, leading to oncogenic transformations.

* Detection & diagnosis :—

Biopsy & Histopathology: A piece of suspected tissue cut into thin sections, stained & examined under a microscope by a pathologist. (Blood/marrow tests used for high cell count in leukemias).

Imaging: Radiography (X-rays). **CT** (Computed Tomography) uses X-rays to generate 3-D internal images. **MRI** uses strong magnetic fields & non-ionising radiations to detect pathological & physiological changes.

Biomarkers & Genetics: Antibodies against cancer-specific antigens. Identification of genes predisposing individuals to cancers helps prevent exposure to specific carcinogens (e.g., tobacco smoke).

* Treatment :—

- Combinations of surgery, radiation therapy & chemotherapy are standard treatments.
- Chemotherapeutic drugs are used to kill cancerous cells, but they have side effects like hair loss and anaemia.
- Patients are given biological response modifiers like α-interferon, which activates their immune system and helps in destroying the tumor.

NotebookLM

Drugs & Alcohol abuse :—

* Opioids :—

Bind to specific opioid receptors present in our CNS & gastrointestinal tract.

E.g.: Heroin (commonly called smack). It is chemically diacetylmorphine (white, odourless, bitter crystalline compound).

Obtained by acetylation of morphine, extracted from latex of poppy plants (Papaver somniferum).

It is a depressant that lowers body functions. Taken by snorting & injection.

* Cannabinoids :—

Act with cannabinoid receptors present in the brain.

Obtained from Cannabis sativa flower.

E.g.: Marijuana, Hashish, Charas & Ganja (produced by flower tops, leaves, or resins).

Taken by oral ingestion or inhalation. They effect the cardiovascular system.

* Coca alkaloid or cocaine :—

Obtained from coca plant or Erythroxylum coca (from SA).

Attacks neurotransmitter dopamine in CNS. Produces a sense of euphoria & increased energy. Causes hallucination.

Taken by snorting. Abused by sportspersons.

Note: Other plants with hallucinogenic properties are Atropa belladonna & Datura. Drugs like barbiturates, amphetamine, benzodiazepines (normally used for mental illness/insomnia) are often abused.

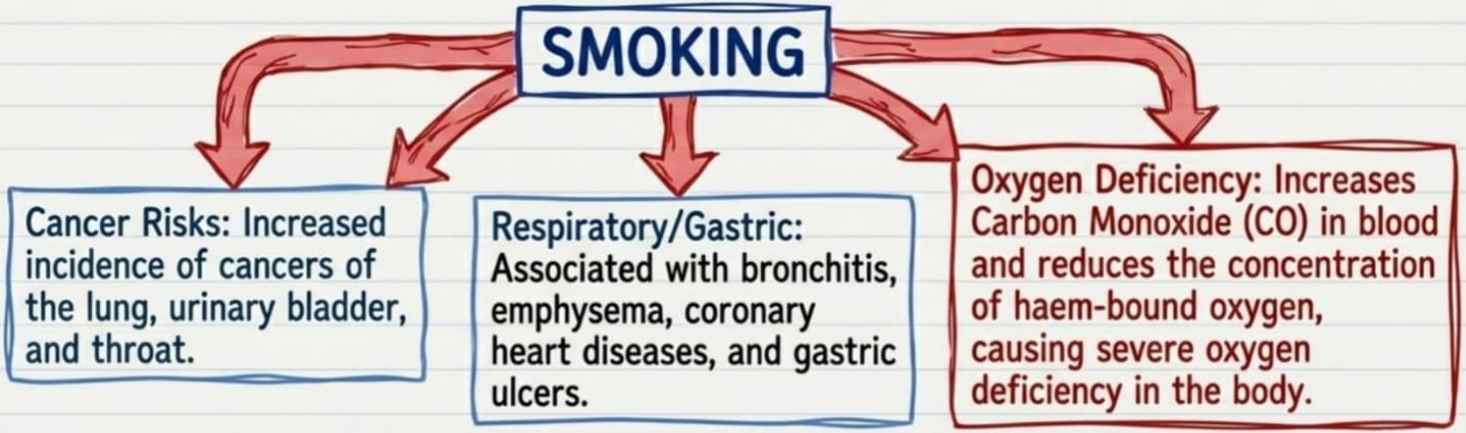
TYPES OF DRUGS	FUNCTION	EXAMPLES	PRODUCED FROM	CHARACTERISTICS
Opioids	Bind to specific opioid receptors in CNS & GI tract	Heroin	Acetylation of morphine, extracted from poppy plants (Papaver somniferum)	Chemically diacetylmorphine, white/odourless/bitter. Depressant that slows body functions.
Cannabinoids	Acts on cannabinoid receptor present in brain	**Marijuana, hashish, charas, ganja**	Flower tops, leaves or resins of Cannabis sativa	Affect cardiovascular system in body.
Coca alkaloid or cocaine	Attacks neurotransmitter dopamine in CNS, produces euphoria & increased energy. Causes hallucination.	**Barbiturates, amphetamine, benzodiazepines etc.**	Obtained from coca plants (Erythroxylum coca)	Normally used as medicines for mental illness but are often abused.
Morphine	**Sedative and painkiller given** to patients undergone surgery			
Tobacco	Stimulates adrenal gland to produce adrenaline, increases BP and heartrate			Increases cancer risks. Smoked, chewed or used as snuff. Contains nicotine.
Smoking	Increases CO in blood and reduces concentration of haem-bound oxygen, causing oxygen deficiency.			Increases risks of cancers (lung, bladder, throat), bronchitis, emphysema, ulcers, heart diseases.

* Morphine :—

It is a sedative & painkiller and is given to patients who have undergone surgery.

* Tobacco & Smoking :—

Tobacco is smoked, chewed, or used as a snuff. It increases risks of cancer of the oral cavity. Contantains nicotine, a type of alkaloid. Stimulates the adrenal gland to produce adrenaline or nor-adrenaline, which raises BP & heartrate.



Master Class Notebook

Adolescence & drug or alcohol abuse :—

Major causes due to which adolescents start:

- Curiosity, need for adventure, excitement & experimentation.
- Perception that it is 'cool' to use drugs, tobacco, cigarette etc.
- Television, movies, newspapers, internet promote this perception.
- Unstable or unsupportive family structures & peer pressure.

Addiction :—

A psychological attachment to certain effects like euphoria & a temporary feeling of well-being. Drives people to take them even when self-destructive.

The Tolerance Cycle: With repeated use, the tolerance level of receptors increases → Receptors respond only to highest doses → Leads to greater intake & addiction.

Dependence :—

The tendency of the body to manifest a characteristic and unpleasant withdrawal syndrome if a regular dose is abruptly discontinued.

Characterized by anxiety, shivering, nausea & sweating.

Prevention & Control :—

- Avoid undue peer pressure.
- Education & counselling.
- Seeking help from parents & peers.
- Looking for danger signs.
- Seeking professional & medical help.

Master Class Notebook

Effects of drugs / alcohol abuse :—

- More prone towards diseases like Hepatitis-B & AIDS due to sharing of syringes.
- Leads to damages to nervous system & liver (cirrhosis).
- Use during pregnancy has an adverse effect on foetus.
- Sportspersons use narcotic analgesics, anabolic steroids, diuretics & hormones to increase muscle strength/bulk & promote aggression to increase athletic performance.

Side-effects in females :—

- Masculinisation.
- Aggression.
- Mood swings & depression.
- Abnormal menstrual cycles.
- Excessive hair growth on face/body.
- Enlargement of clitoris.
- Deepening of voice.

Side-effects in males :—

- Acne.
- Increased aggression.
- Mood swings & depression.
- Reduction of size of testicles.
- Decreased sperm production.
- Potential for kidney & liver dysfunction.
- Breast enlargement.
- Premature baldness.
- Enlargement of prostate gland.